Taller 2. Hands-on session [Docker]

Andres Felipe Cruz

1. Verify Docker Installation

```
PS C:\Users\User> docker version
Client:
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:20:16 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.61.0 (219004)
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:24 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

```
PS C:\Users\User> docker --help
Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  bake        Build from a file
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Authenticate to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  ai*         Docker AI Agent - Ask Gordon
  builder     Manage builds
  buildx*     Docker Buildx
  compose*    Docker Compose
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container
  desktop*    Docker Desktop commands
  extension*  Manages Docker extensions
  image       Manage images
  init*       Creates Docker-related starter files for your project
  manifest    Manage Docker image manifests and manifest lists
  mcp*        Docker MCP Plugin
```

```
For more help on how to use Docker, head to https://docs.docker.com/go/guides/
PS C:\Users\User> docker pull --help
Usage:  docker pull [OPTIONS] NAME[:TAG|@DIGEST]

Download an image from a registry

Aliases:
  docker image pull, docker pull

Options:
  -a, --all-tags          Download all tagged images in the repository
      --platform string   Set platform if server is multi-platform capable
  -q, --quiet             Suppress verbose output
PS C:\Users\User>
```

1. ¿Cuántos argumentos son absolutamente requeridos por docker pull?
   r/ Necesita un argumento obligatorio, que sería el nombre de la imagen.

2. ¿Qué es un registry?

r/ Es el lugar desde donde Docker descarga (pull) las imágenes y donde también pueden subirse (push) imágenes creadas por los usuarios

```
PS C:\Users\User> docker pull japeto/pujgcc:latest
latest: Pulling from japeto/pujgcc
f67323742a64: Pull complete
1efbd2d5674a: Pull complete
15a33158a136: Pull complete
aa18ad1a0d33: Pull complete
c6dbbe32db7e: Pull complete
6461d3be7619: Pull complete
8287aab02fcd: Pull complete
c4b45e832c38: Pull complete
edd3d1d6fad6: Pull complete
e5d4afe2cf59: Pull complete
022a58c161f7: Pull complete
Digest: sha256:9b3d7bbf410396d0a26e6bcffbb54e4239736d5a23ad694b03d93c26f9fc6868
Status: Downloaded newer image for japeto/pujgcc:latest
docker.io/japeto/pujgcc:latest
PS C:\Users\User> docker images

IMAGE                  ID            DISK USAGE   CONTENT SIZE   EXTRA
japeto/pujgcc:latest   9b3d7bbf4103       2.18GB         448MB
japeto/so-tools:v64    7e990f123c8d          2GB         506MB    U
ubuntu:24.04           cd1dba651b30        119MB        31.7MB    U
PS C:\Users\User>
```

3. ¿Cuál es el tamaño de la imagen?

   r/ el tamaño de la imagen es de 2.1GB, pero el contenido es de 448MB

4. ¿Cuál es el Image ID?

   r/ El ID que se le asigno a esta imagen es: 9b3d7bbf4103

```
PS C:\Users\User> docker inspect japeto/pujgcc:latest
[
    {
        "Id": "sha256:9b3d7bbf410396d0a26e6bcffbb54e4239736d5a23ad694b03d93c26f9fc6868",
        "RepoTags": [
            "japeto/pujgcc:latest"
        ],
        "RepoDigests": [
            "japeto/pujgcc@sha256:9b3d7bbf410396d0a26e6bcffbb54e4239736d5a23ad694b03d93c26f9fc6868"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2023-02-08T03:10:14.416761058Z",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "GPG_KEYS=B215C1633BCA0477615F1B35A5B3A004745C015A \tB3C42148A44E6983B3E4CC0793FA9B1AB75C61B8 \t90AA470469
97C103468EE5D750B583AB00996FC26A641 \t33C235A34C46AA3FFB293709A328C3A2C3C45C06",
                "GCC_VERSION=4.9.4"
            ],
            "Cmd": [
                "bash"
            ],
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 448257041,
                "sha256:915743e74481c848f930b5be010696fa914adae107ffa8177ae7d7d88dddfa0b",
                "sha256:64a82172c2221b36c9c0b5e8e7eb8c46608d30cdad661f984713012a2b686ea1",
                "sha256:72821cda15288575f59f47376e3ea5543f0ec378701993bdf4588e5350e85384",
                "sha256:e178c0887daafcd99439ab2f5371cf47b282342c45f5c4650c38a77330625089",
                "sha256:94da6388edf5d61e19db659eee6aeeccac02a42eab3dd78755631477e38afee8"
            ]
```

5. ¿Arquitectura CPU?

   r/ la arquitectura es amd64

6. ¿Sistema operativo?

   r/ el sistema operativo de esta imagen es Linux

```
PS C:\Users\User> docker run --help
Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

Aliases:
  docker container run, docker run

Options:
      --add-host list                    Add a custom host-to-IP mapping
                                         (host:ip)
      --annotation map                   Add an annotation to the
                                         container (passed through to the
                                         OCI runtime) (default map[])
  -a, --attach list                      Attach to STDIN, STDOUT or STDERR
      --blkio-weight uint16              Block IO (relative weight),
                                         between 10 and 1000, or 0 to
                                         disable (default 0)
      --blkio-weight-device list         Block IO weight (relative device
                                         weight) (default [])
      --cap-add list                     Add Linux capabilities
      --cap-drop list                    Drop Linux capabilities
      --cgroup-parent string             Optional parent cgroup for the
                                         container
      --cgroupns string                  Cgroup namespace to use
                                         (host|private)
                                         'host':    Run the container in
                                         the Docker host's cgroup
                                         namespace
                                         'private': Run the container in
                                         its own private cgroup namespace
                                         '':        Use the cgroup
                                         namespace as configured by the
                                                    default-cgroupns-mode
```

7. ¿Cuáles son las opciones y parámetros del comando run?
   r/ Los principales podrían ser -it, -d, -rm, -v, -p, --name, --network, etc. Este comando dio muchas opciones.
8. ¿cuál es la diferencia entre una imagen y un contenedor?
   r/ la imagen seria como una plantilla y el contenedor una instancia de dicha plantilla

```
PS C:\Users\User> docker run japeto/pujgcc bash --help
GNU bash, version 4.3.30(1)-release-(x86_64-pc-linux-gnu)
Usage:  bash [GNU long option] [option] ...
        bash [GNU long option] [option] script-file ...
GNU long options:
        --debug
        --debugger
        --dump-po-strings
        --dump-strings
        --help
        --init-file
        --login
        --noediting
        --noprofile
        --norc
        --posix
        --rcfile
        --restricted
        --verbose
        --version
Shell options:
        -ilrsD or -c command or -O shopt_option         (invocation only)
        -abefhkmnptuvxBCHP or -o option
Type `bash -c "help set"' for more information about shell options.
Type `bash -c help' for more information about shell builtin commands.
Use the `bashbug' command to report bugs.
PS C:\Users\User>
```

9. ¿Qué se mostró? ¿Docker o bash?
   r/ la respuesta que se mostró provenía del bash.

```
PS C:\Users\User> docker run -it japeto/pujgcc bash
root@f3d64e267590:/# pwd
/
root@f3d64e267590:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@f3d64e267590:/# ls /home
root@f3d64e267590:/# pwd
/
root@f3d64e267590:/#
```

10. ¿Cuál es la dirección actual?

    r/ el directorio actual es /. Este debe ser el root del container

11. ¿Cuántos archivos hay en el directorio actual?

    r/ luego de ejecutar el comando ls podemos observar que hay bastantes directorios del sistema.

12. ¿Existe la carpeta /home dentro del contenedor?

    r/ si, pero está vacía

```
PS C:\Users\User> docker inspect --format='{{.Config.Volumes}}' japeto/pujgcc:latest
map[]
PS C:\Users\User> docker run japeto/pujgcc:latest ls /home
PS C:\Users\User>
```

```
PS C:\Users\User> docker run -v ${PWD}:/home/ japeto/pujgcc:latest ls /home/
AppData
Cisco Packet Tracer 8.2.2
Configuración local
Contacts
Cookies
Datos de programa
Desktop
Documents
Downloads
Entorno de red
Favorites
Impresoras
Links
Menú Inicio
Mis documentos
Music
NTUSER.DAT
NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TM.blf
NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TMContainer00000000000000000001.regtrans-ms
NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TMContainer00000000000000000002.regtrans-ms
OneDrive
Pictures
Plantillas
Reciente
Saved Games
Searches
SendTo
Videos
active_bets.json
```

```
PS C:\Users\User> docker run -it -v ${PWD}:/home/ japeto/pujgcc:latest
root@dd04205a42d7:/# cd /home
root@dd04205a42d7:/home# nano fibonacci.cpp
root@dd04205a42d7:/home# pwd
/home
root@dd04205a42d7:/home# ls
AppData                 Downloads      NTUSER.DAT                                                                          Saved Games        ntuser.dat.LOG1
Cisco Packet Tracer 8.2.2  Entorno de red  NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TM.blf                          Searches           ntuser.dat.LOG2
Configuraci??n local    Favorites      NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TMContainer00000000000000000001.regtrans-ms  SendTo             ntuser.ini
Contacts                Impresoras     NTUSER.DAT{00a3c595-80ad-11f0-bdb3-d841c0ecbe7b}.TMContainer00000000000000000002.regtrans-ms  Videos             source
Cookies                 Links          OneDrive                                                                            active_bets.json
Datos de programa       Men?? Inicio   Pictures                                                                            ansel
Desktop                 Mis documentos Plantillas                                                                          fibonacci.cpp
Documents               Music          Reciente                                                                            import requests.py
root@dd04205a42d7:/home# g++ fibonacci.cpp -o fibo
root@dd04205a42d7:/home# ./fibo
Ingrese el número de términos: 65
Serie Fibonacci: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 241
57817 39088169 63245986 102334155 165580141 267914296 433494437 701408733 1134903170 1836311903 -1323752223 512559680 -811192543 -298632863 -1109825406 -1408458269 1776683621 368225352 2144908973 -178183
2971 363076002 -1418756969 -1055680967 1820529360 764848393 -1709589543 -944741150 1640636603 root@dd04205a42d7:/home#
```

Se ha creado un archivo dentro del container, luego lo editamos para que tenga la lógica de Fibonacci para luego ejecutarlo.

```
PS C:\Users\User> docker run --rm -v ${PWD}:/home/ japeto/pujgcc:latest g++ -o /home/facto /home/factorial.cpp
PS C:\Users\User> docker run --rm -v ${PWD}:/home/ japeto/pujgcc:latest /home/facto
120
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 5;
    int fact = 1;

    for(int i = 1; i <= n; i++)
        fact *= i;

    cout << fact;
    return 0;
}
```

Ejemplo de creación de factorial en mi máquina. Compilación y ejecución en el container.

```
PS C:\Users\User> docker run -it --name tcontainer japeto/pujgcc:latest /bin/bash
root@6aec2cd294f5:/# apt update
Ign http://security.debian.org jessie/updates InRelease
Ign http://security.debian.org jessie/updates Release.gpg
Ign http://security.debian.org jessie/updates Release
Err http://security.debian.org jessie/updates/main amd64 Packages

Err http://security.debian.org jessie/updates/main amd64 Packages
Ign http://deb.debian.org jessie Release
Ign http://deb.debian.org jessie-updates Release
Err http://deb.debian.org jessie/main amd64 Packages
  404  Not Found [IP: 199.232.178.132 80]
Err http://deb.debian.org jessie-updates/main amd64 Packages
  404  Not Found [IP: 199.232.178.132 80]
W: Failed to fetch http://security.debian.org/dists/jessie/updates/main/binary-amd64/Packages  404  Not Found [IP: 151.101.66.132 80]

W: Failed to fetch http://deb.debian.org/debian/dists/jessie/main/binary-amd64/Packages  404  Not Found [IP: 199.232.178.132 80]

W: Failed to fetch http://deb.debian.org/debian/dists/jessie-updates/main/binary-amd64/Packages  404  Not Found [IP: 199.232.178.132 80]

E: Some index files failed to download. They have been ignored, or old ones used instead.
root@6aec2cd294f5:/# apt install python3
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  dh-python libmpdec2 libpython3-stdlib libpython3.4-minimal libpython3.4-stdlib python3-minimal python3.4 python3.4-minimal
Suggested packages:
  python3-doc python3-tk python3-venv python3.4-venv python3.4-doc binfmt-support
The following NEW packages will be installed:
  dh-python libmpdec2 libpython3-stdlib libpython3.4-minimal libpython3.4-stdlib python3 python3-minimal python3.4 python3.4-minimal
0 upgraded, 9 newly installed, 0 to remove and 166 not upgraded.
Need to get 4667 kB of archives.
After this operation, 18.8 MB of additional disk space will be used.
```

13. ¿Espacio adicional?

    r/ luego de ejecutar ambos comandos nos dice que 18.8 MB adicionales en el disk se han usado.

14. ¿Versión instalada?

    r/ se instaló la versión 2.7.9

```
root@6aec2cd294f5:/# apt install vim
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  libgpm2 vim-runtime
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libgpm2 vim vim-runtime
0 upgraded, 3 newly installed, 0 to remove and 166 not upgraded.
Need to get 6038 kB of archives.
After this operation, 28.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
WARNING: The following packages cannot be authenticated!
  libgpm2 vim-runtime vim
Install these packages without verification? [y/N] y
Err http://deb.debian.org/debian/ jessie/main libgpm2 amd64 1.20.4-6.1+b2
  404  Not Found [IP: 199.232.178.132 80]
Err http://security.debian.org/ jessie/updates/main vim-runtime all 2:7.4.488-7+deb8u4
  404  Not Found [IP: 151.101.130.132 80]
Err http://security.debian.org/ jessie/updates/main vim amd64 2:7.4.488-7+deb8u4
  404  Not Found [IP: 151.101.130.132 80]
E: Failed to fetch http://deb.debian.org/debian/pool/main/g/gpm/libgpm2_1.20.4-6.1+b2_amd64.deb  404  Not Found [IP: 199.232.178.132 80]

E: Failed to fetch http://security.debian.org/pool/updates/main/v/vim/vim-runtime_7.4.488-7+deb8u4_all.deb  404  Not Found [IP: 151.101.130.132 80]

E: Failed to fetch http://security.debian.org/pool/updates/main/v/vim/vim_7.4.488-7+deb8u4_amd64.deb  404  Not Found [IP: 151.101.130.132 80]

E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
root@6aec2cd294f5:/#
```

```
root@6aec2cd294f5:/# exit
exit
PS C:\Users\User> docker ps -l
CONTAINER ID   IMAGE                 COMMAND       CREATED         STATUS                      PORTS     NAMES
6aec2cd294f5   japeto/pujgcc:latest  "/bin/bash"   7 minutes ago   Exited (100) 32 seconds ago           tcontainer
PS C:\Users\User>
```

15. ¿Qué sucede al salir?

    r/ El contenedor queda detenido

```
PS C:\Users\User> docker commit tcontainer japeto/tcontainer:1.0
PS C:\Users\User> docker images

IMAGE                   ID             DISK USAGE   CONTENT SIZE   EXTRA
japeto/pujgcc:latest    9b3d7bbf4103       2.18GB          448MB    U
japeto/so-tools:v64     7e990f123c8d          2GB          506MB    U
japeto/tcontainer:1.0   a8b9edd6f97c       2.18GB          448MB
ubuntu:24.04            cd1dba651b30        119MB         31.7MB    U
PS C:\Users\User> docker rmi japeto/tcontainer:1.0
Untagged: japeto/tcontainer:1.0
Deleted: sha256:a8b9edd6f97c8ba92a2cea4d1d782f22a8cf00af76db7d814cdd72023ddb94a3
PS C:\Users\User> docker images

IMAGE                   ID             DISK USAGE   CONTENT SIZE   EXTRA
japeto/pujgcc:latest    9b3d7bbf4103       2.18GB          448MB    U
japeto/so-tools:v64     7e990f123c8d          2GB          506MB    U
ubuntu:24.04            cd1dba651b30        119MB         31.7MB    U
```

```
PS C:\Users\User> docker pull gcc
Using default tag: latest
latest: Pulling from library/gcc
aa605359ee41: Pull complete
be86affbdf30: Pull complete
128c71264009: Pull complete
49a31cdc1eab: Pull complete
bd637997358e: Pull complete
ef235bf1a09a: Pull complete
954d6059ca7b: Pull complete
b5e2021c4c8b: Pull complete
cd1d9ab87452: Download complete
f0f91c983fc0: Download complete
Digest: sha256:0cc1665930da5d3052d4af019579f1f35dbdf02598ddadfba3b15365b0507de5
Status: Downloaded newer image for gcc:latest
docker.io/library/gcc:latest
```

Hemos creado una nueva imagen con base a la que ya teníamos, pero esto puede suponer algunos problemas y/o complicación, por lo tanto la eliminamos y bajamos la imagen base de gcc, para luego crear nuestro dockerfile para poder hacer el build de la imagen y un programa simple pa a ejecutarlo.

```
PS C:\Users\User\sorting> ls
PS C:\Users\User\sorting> ls


    Directorio: C:\Users\User\sorting


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        19/02/2026      4:23            105 Dockerfile.dockerfile
-a----        19/02/2026      4:24            412 main.c


PS C:\Users\User\sorting>
```

```
PS C:\Users\User\sorting> docker run --rm japeto/sorting
1 2 3 5 9
PS C:\Users\User\sorting>
```

Luego de ejecutar el build de esta nueva imagen teniendo en base la de gcc, logramos crear nuestro dockerfile y ejecutar el programa.